PaperSonar

Business requirements

Design baseline 1.0 / 25 October 2026

This document defines the decisions, data coverage and acceptance conditions for PaperSonar's initial analytical release. Requirements use stable identifiers so reports, tables and verification work can be traced to the intended behavior.

Stakeholders and decisions

User	Decision supported
Researcher	Which of several directions warrants a closer literature review?
Supervisor or research lead	What evidence supports a proposed direction and what limits the comparison?
Research group	Which institutions, authors and venues should be investigated?
Data maintainer	Is a refreshed corpus complete enough to publish without altering prior evidence?

Business requirements

ID	Requirement	Acceptance
BR01	Compare research activity over completed years.	Q01 and Q02 return counts, denominators and declared coverage.
BR02	Compare citation measures fairly.	Q03 groups by publication year and type at a single observation date.
BR03	Investigate potential topic intersections.	Q05 and Q13 show source papers, parent volumes and counting-method sensitivity.
BR04	Explore scholarly participation.	Q06 to Q11 show attributable records and distinguish fractional from full counts.
BR05	Retain the evidence behind a comparison.	A dossier identifies corpus, extraction window and mapping and metric versions.
BR06	Keep incomplete or unreliable comparisons visible.	Q12 and Q14 label missingness, partial years and unavailable observations.

Functional and data requirements

ID	Required behavior
FR01	Read a versioned corpus configuration with topic IDs, years, admitted types and resource limits.
FR02	Extract cursor pages within the free usage budget. Retain raw files, checksums, source URLs with secrets redacted and checkpoints.
FR03	Stage source records and quarantine malformed rows with a work ID or source page reference and reason.
FR04	Resolve canonical OpenAlex IDs and normalized DOIs. Load dimensions, facts and complete attribution sets without multiplying work counts.
FR05	Publish observations only after a complete successful corpus refresh. Preserve prior dates and replace a same-day observation atomically.
FR06	Refresh the bounded corpus without depending on paid created-date or updated-date sync filters.
FR07	Provide approved SQL measures for fourteen questions and five report pages with record drill-through.
FR08	Export a comparison with corpus version, observation date, cutoff, counting method and supporting identifiers.
FR09	Keep reference-only stubs separate from admitted publications. Retain first local observation of a citation pair.
FR10	Record missing fields separately from measured zero values and preserve revision flags in comparative views.

A valid admitted publication has a canonical OpenAlex work ID and satisfies the configured topic, year and type criteria. Title, venue, DOI, author and institution fields may be missing. Missing descriptive dimensions use key 0. A missing citation count remains null. Source-reported publication dates are not evidence of day-level precision, so elapsed-period reports exclude year-only and unknown dates.

The analytical population includes every retrieved admitted work after canonical identity reconciliation. Reference targets outside that population may exist as dim_work stubs but have no publication fact. The pilot does not define the final population. Its purpose is to profile fields and check source access.

Quality requirements and acceptance

ID	Acceptance condition
NFR01	A replay of the same successful raw input changes neither admitted work count nor citation pair count. All primary and foreign keys remain valid.
NFR02	Distinct raw admitted works equal loaded publication facts plus documented rejected or unresolved records. Overlapping source partitions are reconciled before this check.
NFR03	Every fact row and observation links to a pipeline run. That run resolves to raw manifests and a versioned configuration.
NFR04	A failed refresh leaves the last successful reporting state available. No partially loaded run is selected as the latest snapshot.
NFR05	Fractional weights sum to 1 per work within each populated topic or institution bridge, within tolerance 0.000001. Missing sets are reported separately.
NFR06	Each core report query has a target runtime below five seconds on the declared local test corpus. This is a target to benchmark, not a verified result.
NFR07	Extraction processes pages sequentially and stops for review at the configured 2 GB database or 1 GB compressed raw budget.
NFR08	Keys and credentials remain outside source control and exports. Database credentials are read-only for reporting. SQL parameters and escaped text are required.
NFR09	A clean database can be rebuilt from retained inputs, configuration and scripts. A sample subset supports a small reproducible run.
NFR10	All visual totals reconcile with the approved SQL views, including unknown categories and one selected snapshot date.

Assumptions and dependencies

The bounded source corpus must fit the free API allowance and local storage. Full cursor extraction requires a free OpenAlex key. Crossref and ROR are supplementary sources and can be retried without corrupting the core load. Public user accounts, hosted dashboards and subscription billing are deferred. Model and metric changes require versioned report exports before replacing current metadata.

Approval baseline

The model and requirements are proposed design specifications. Implementation is accepted when all listed conditions have been demonstrated against the declared corpus. Market demand, runtime targets and analytical findings have not yet been established.